

The Hong Kong Polytechnic University

Department of Electronic and Information Engineering

EIE2111 Lab 5: Functions and an Introduction to Recursion

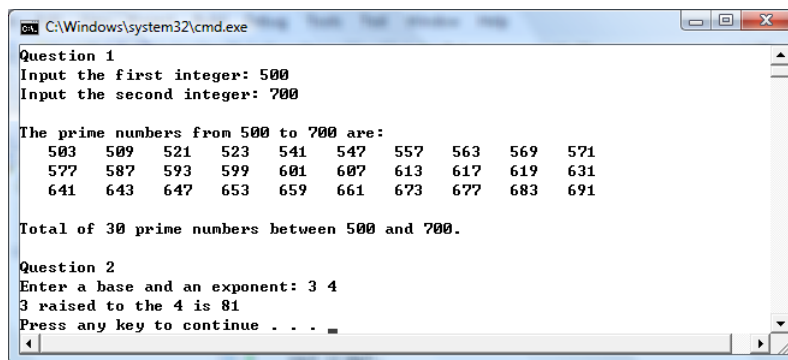
(Deadline for Submission: Check the course information)

Introduction

This laboratory exercise is designed to give you hand-on experience in Functions and an Introduction to Recursion.

Questions

1. An integer is said to be *prime* if it is divisible by only 1 and itself. For example, 2, 3, 5 and 7 are prime, but 4, 6, 8 and 9 are not. Write a function that determines whether a number is prime. You should also write a testing program to prompt users to input two positive integers and find out all prime numbers in between these two integers by using the above function.
2. Write a recursive function `power( a, b )` that, when invoked, returns a^b . For example, `power( 3, 4 ) = 3 × 3 × 3 × 3`. Assume that `b` is an integer greater than or equal to 1. [Hint: The recursion step would use the relationship $a^b = a \times a^{b-1}$ and the terminating condition occurs when `b` is equal to 1.]
3. Combine the above two programs into one. The following is the sample output:



```
C:\Windows\system32\cmd.exe
Question 1
Input the first integer: 500
Input the second integer: 700

The prime numbers from 500 to 700 are:
503 509 521 523 541 547 557 563 569 571
577 587 593 599 601 607 613 617 619 631
641 643 647 653 659 661 673 677 683 691

Total of 30 prime numbers between 500 and 700.

Question 2
Enter a base and an exponent: 3 4
3 raised to the 4 is 81
Press any key to continue . . .
```

Instructions

- a. You are required to submit your C++ programs (the whole projects created in Microsoft Visual Studio 2019) in Question 3 to Blackboard. Zip all of them into a single file.
- b. The deadline of the submission: Check the course information.
- c. It is not required to create any classes or header files for the above exercises. It is fine if all program codes are in the main program.

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